Course Registration System using Composite Design Pattern

- **Problem:** Traditional systems treat courses as flat structures
- **Target Users:** Students & Academic Administrators

System Design

- Hierarchical architecture (Course → Module → SubModule)
- Key Classes:
  - CourseComponent (abstract)
  - Course (composite)
  - Module / SubModule (leaf)
  - Student
  - RegistrationManager
- Package Structure:
  - model
  - service
  - util

OOP Concepts Used

Abstraction

CourseComponent

Inheritance

Course & Module extend base class

Polymorphism

Same method, different behavior

Design Pattern

Composite Pattern

Technical Highlight

- Uniform handling of composite & leaf nodes
- Recursive structure for modules
- Flexible and scalable design

Live Demo

01

Add course and modules

02

Register student

03

Display structure

04

Error handling